

NPTEL Course on
"Computer System Optimization for ML"

Lecture #23

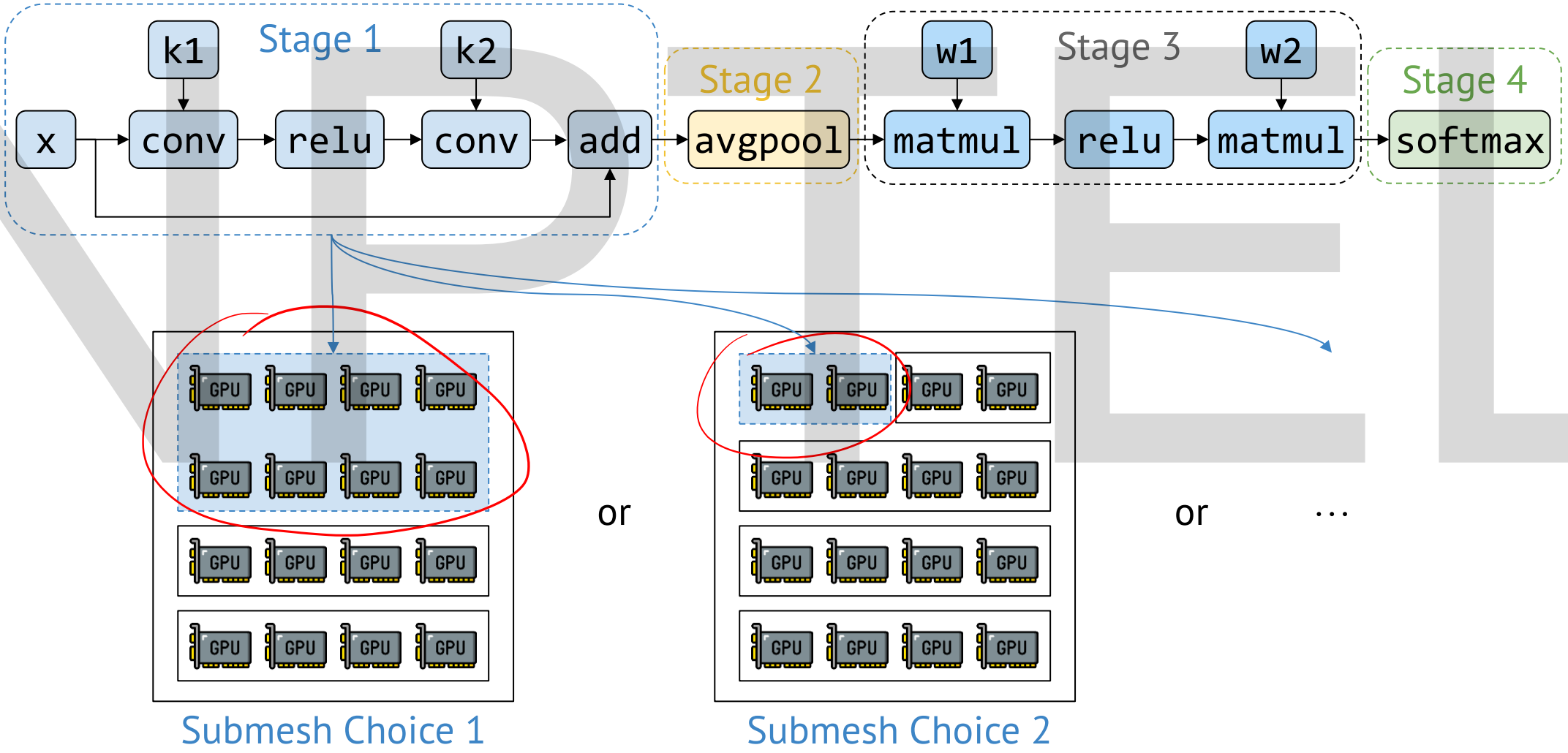
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Agenda

- Recap
- Alpa (contd.)

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Inter-Op Pass



Inter-op Pass: Dynamic Programming Formulation

$$\text{Pipeline execution latency} = \text{Latency of a single datapoint through all stages} + \text{Max stage latency} \times (\#\text{Batch} - 1)$$

Goal: Derive the best *graph partition* and *device assignment* by selecting the optimal set of (stage, submesh) pairs that minimizes this latency.

Input: Run intra-op pass and profile for all possible (stage, submesh) pairs.

Algorithm: Enumerate all possible second term and convert the first term into a 2-dimensional knapsack problem.

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Agenda

- Recap
- **Alpa (contd.)**

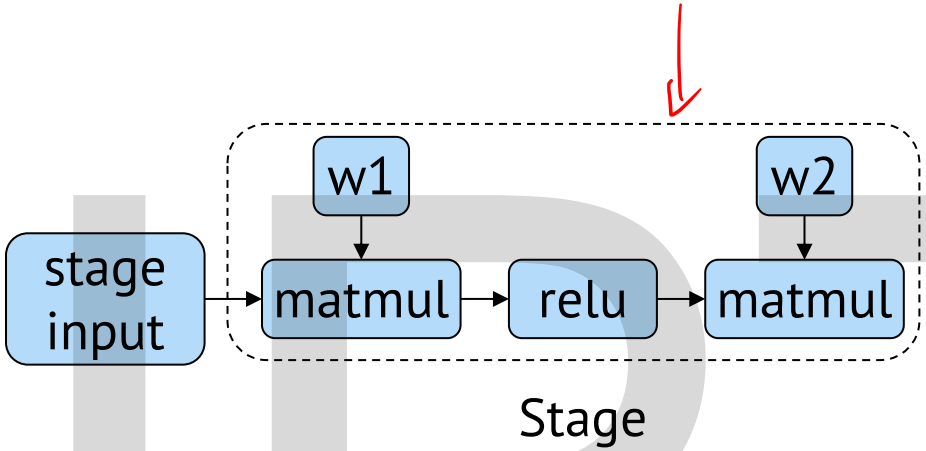
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DP Assumptions

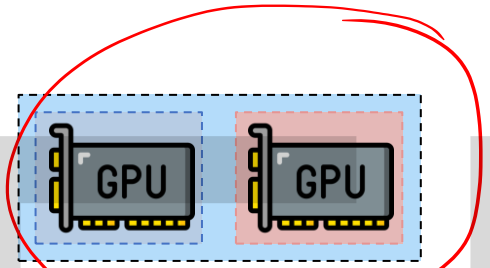
- The forward and the backward pass of the same substage are assigned to the same sub-mesh and considered together
- For a cluster of shape (N, M), only choose the following sub-mesh shape
  - Within a node: (1,1), (1,2), (1,4), (1, 8), ……, (1, M)
  - Cross full nodes: (2, M), (3, M), …., (N, M)
- Set the cost of all 'out of memory' setting to be infinity in DP
- DP complexity: $O\left(K^5 NM(N + log\ M)\right)$
  - K is the number of operators in the graph
  - (N, M) is the mesh shape

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Intra-op Pass

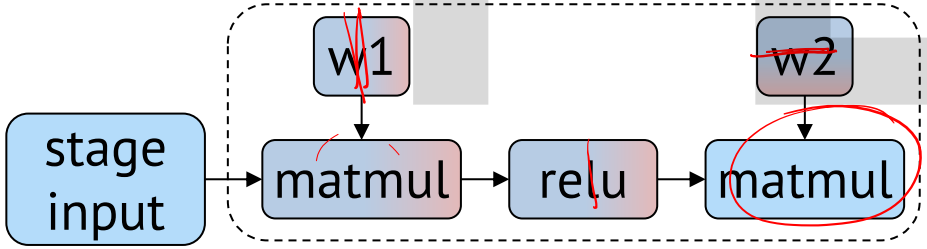


Stage

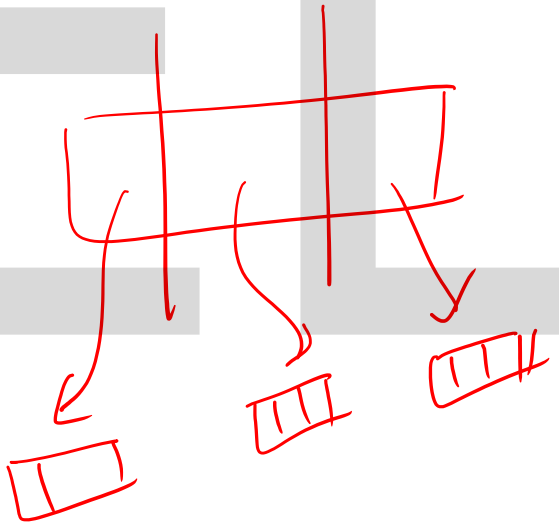


Submesh

Solved by **Integer Linear Programming**



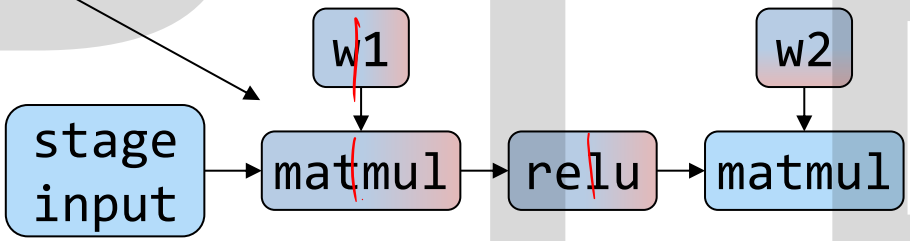
Stage with intra-operator parallelization



Intra-op Pass

- **Integer Linear Programming Formulation**

Decision vector
Parallel strategies of each operator

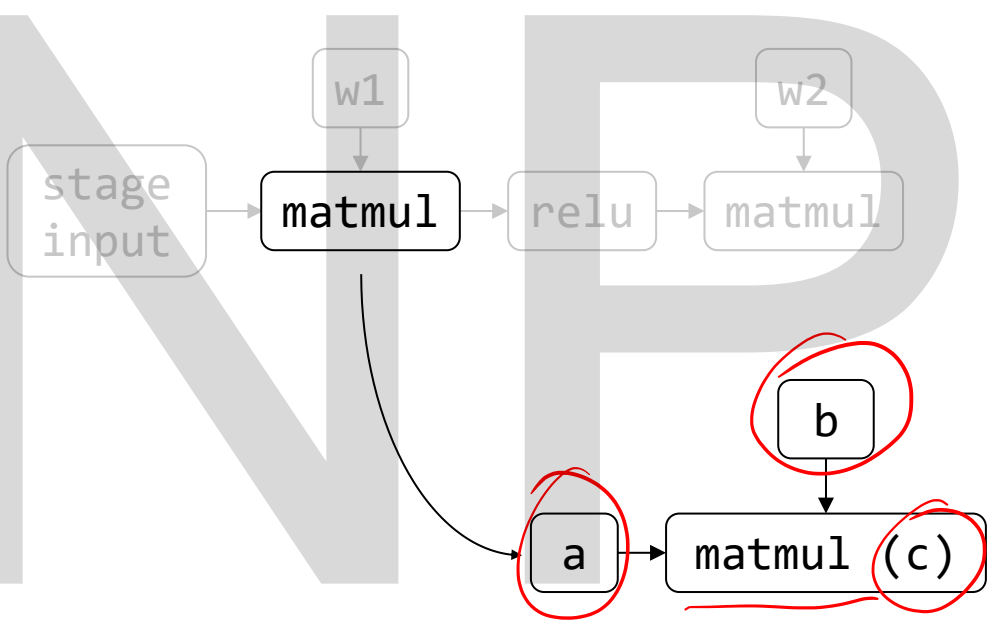


Minimize Computation cost + Communication cost

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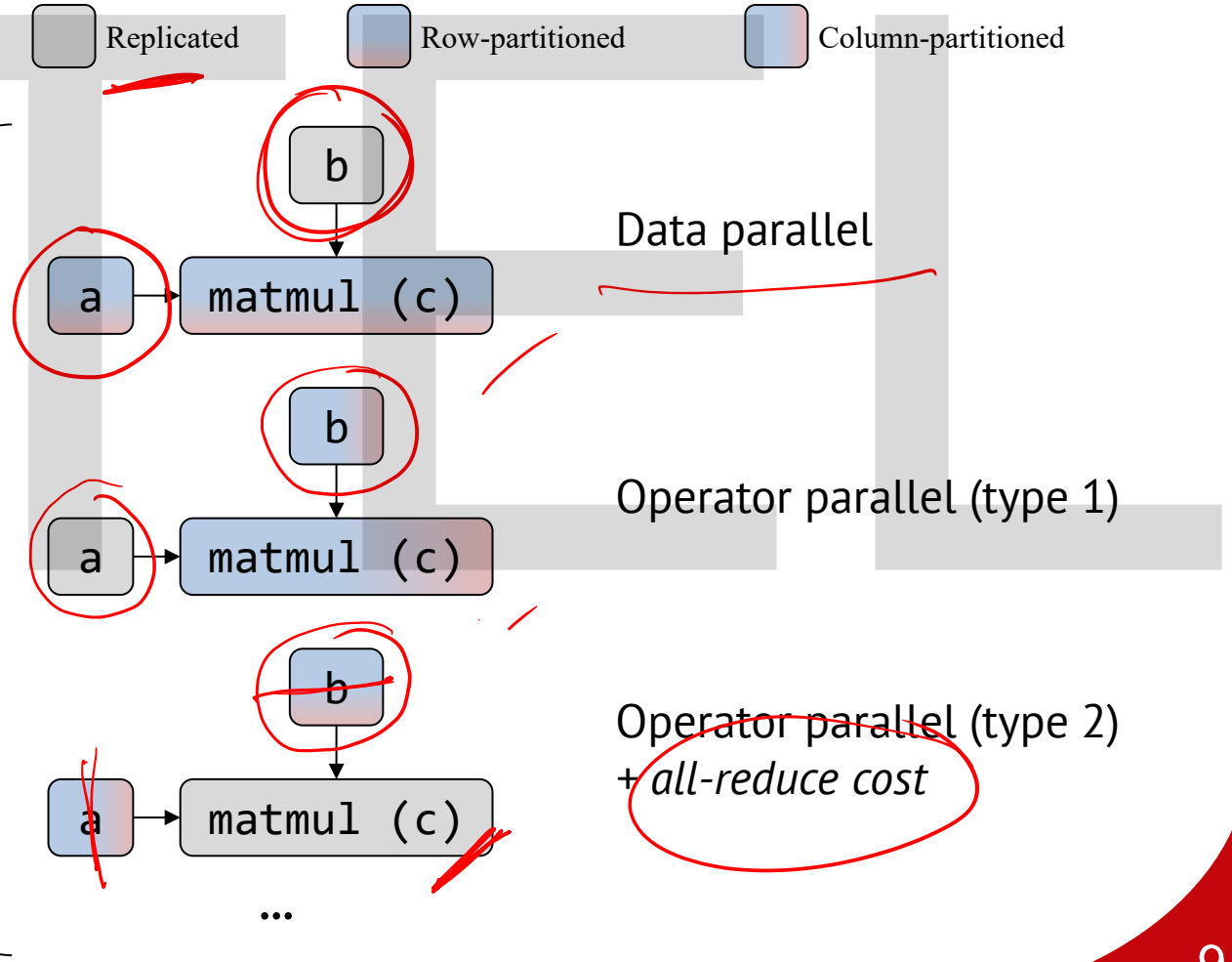
Operator Parallelization Strategy

- **Alpa defines a set of parallelization strategy for each operator (e.g., 2D matmul)**



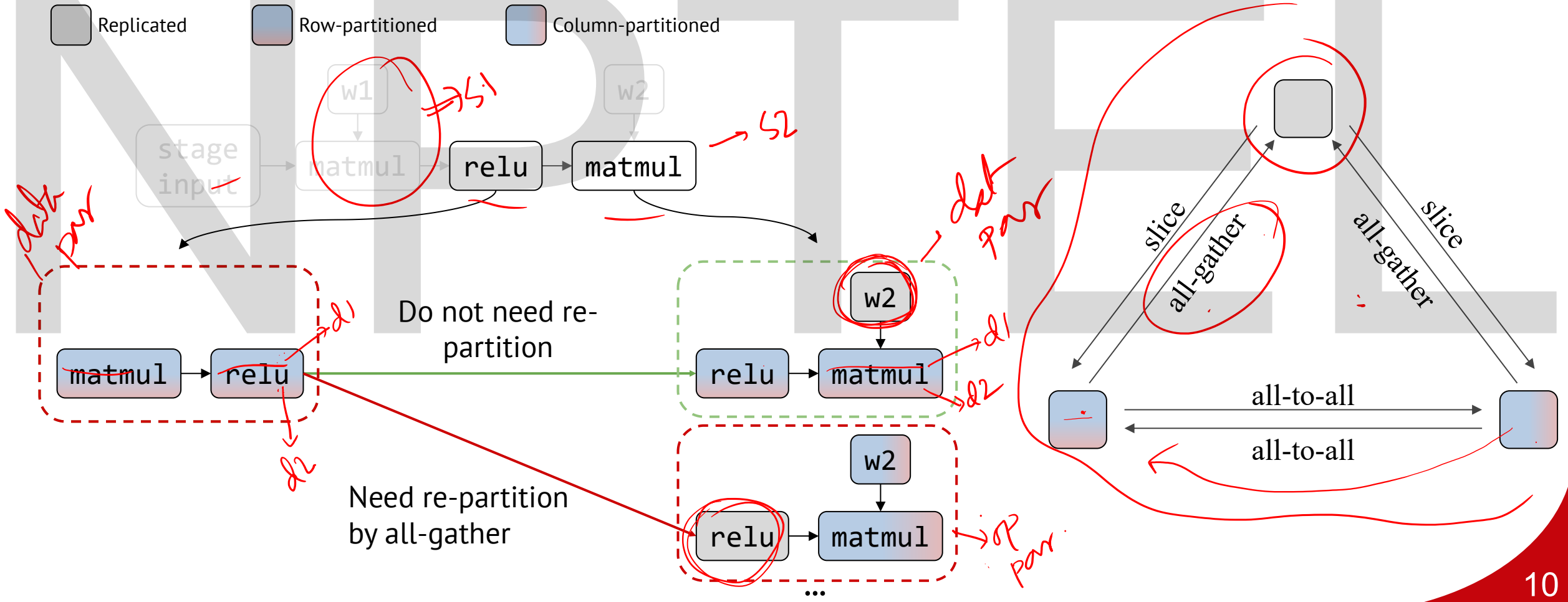
$c[i, j] = \Sigma_k\ a[i, k] \cdot b[k, j]$

Single operator cost



Re-partition Communication Cost

- **Different operator's parallelization strategies might require different partitions of the same tensor**



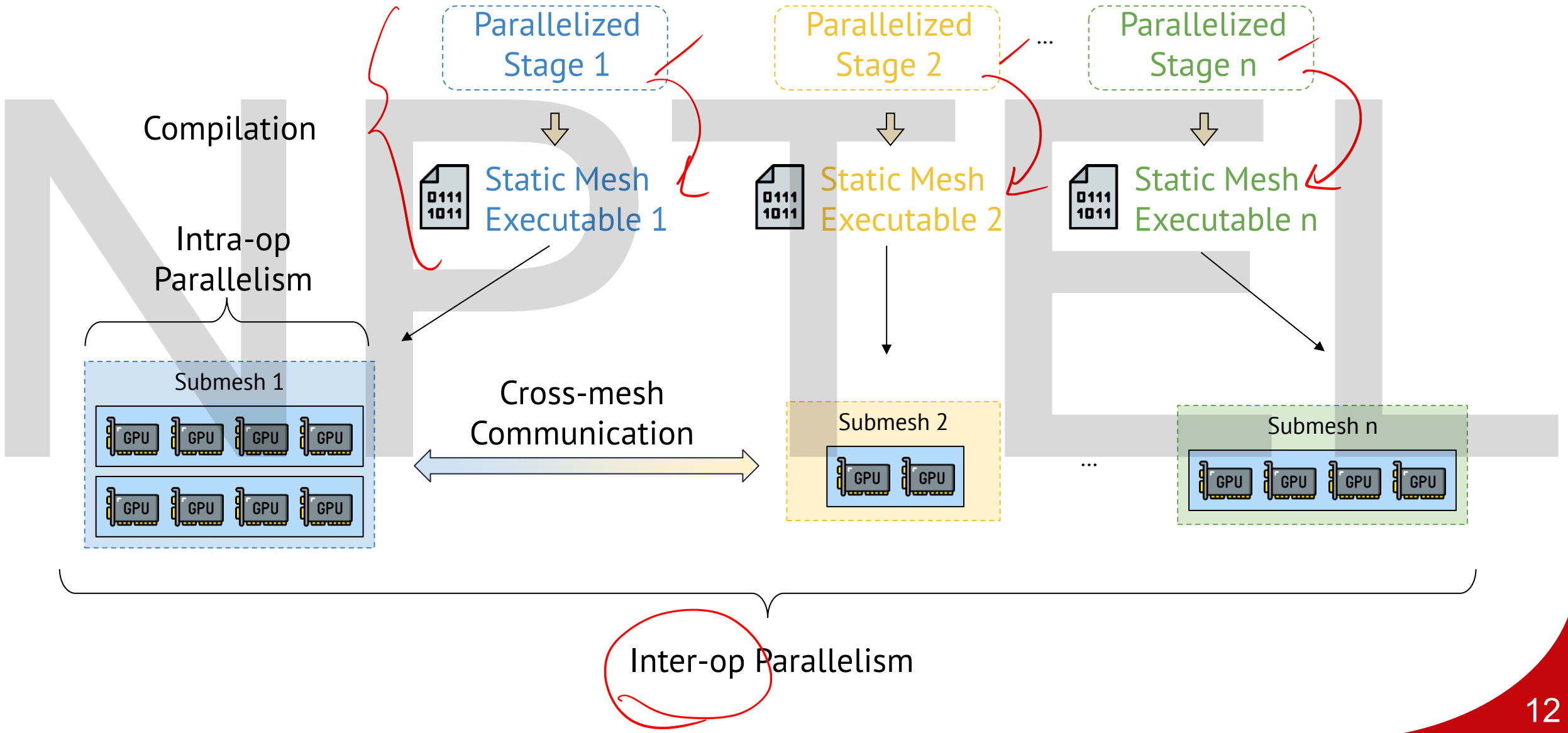
ILP Formulation

- Every operator : define a one-hot decision vector $s_v \in \{0,1\}$ to represent the algorithm it uses. $s_{vi} = 1$ means we pick the ith algorithm for node v.
- c_{vi} is the communication cost of the ith algorithm
- Node v has a compute cost vector d_v
- R – resharding cost

$$\min_{s} \sum_{v \in V} s_v^T (c_v + d_v) + \sum_{(v,u) \in E} s_v^T R_{vu} s_u,$$

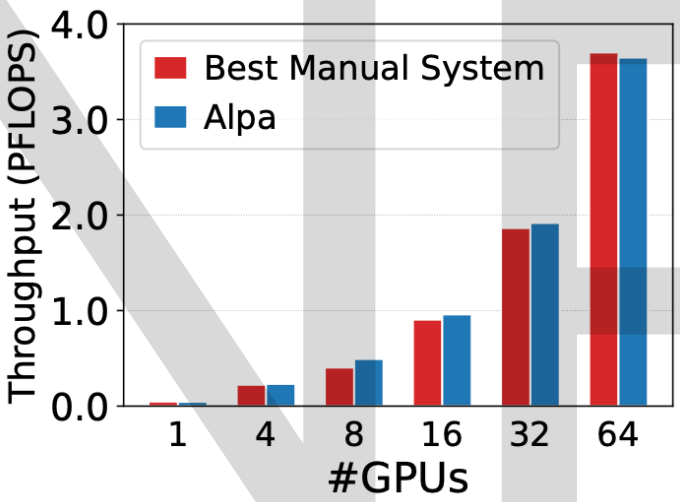
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Runtime Orchestration



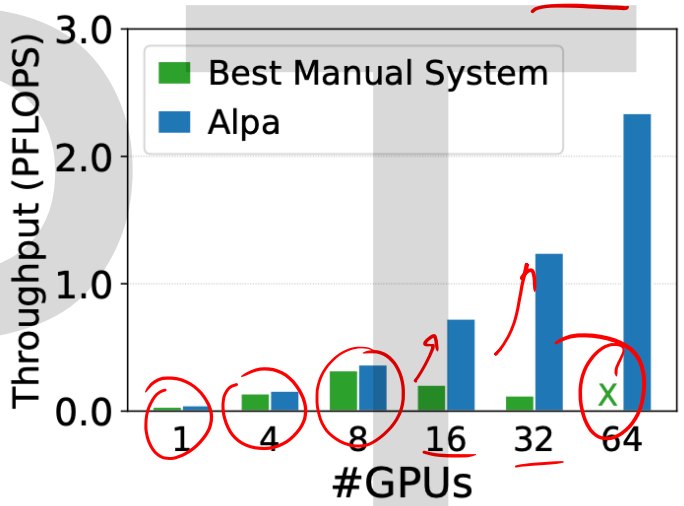
Experimental Evaluation

GPT (up to 39B)



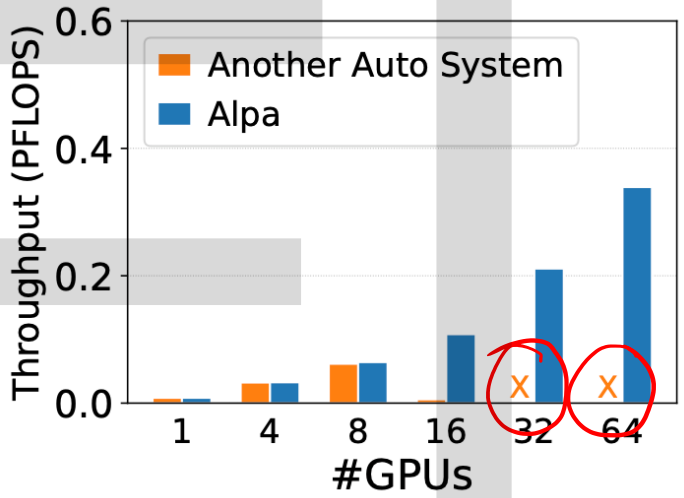
Match specialized manual systems.

GShard MoE (up to 70B)



Outperform the manual baseline by up to 8x.

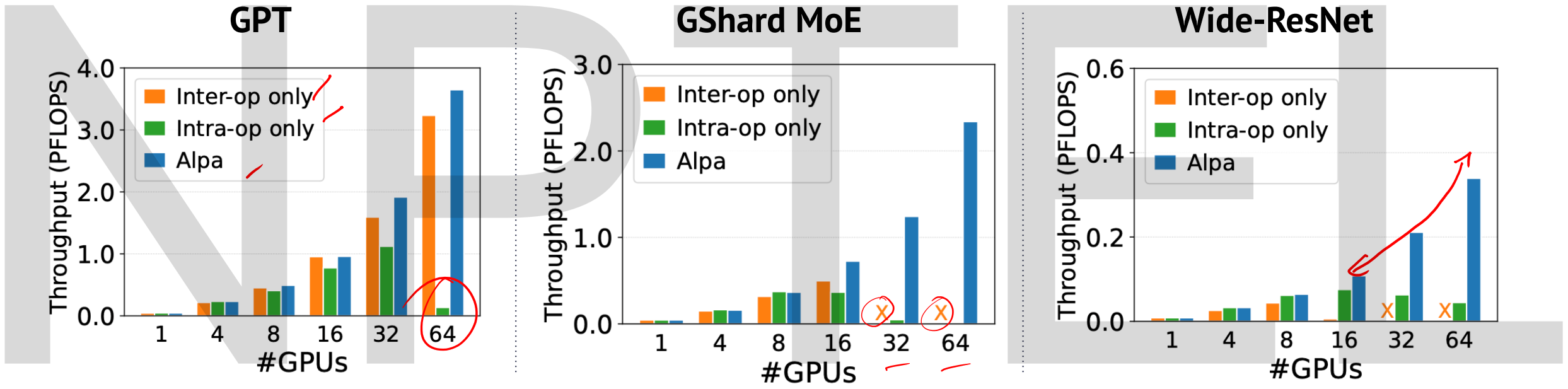
Wide-ResNet (up to 13B)



Generalize to models without manual plans.

Weak scaling results where the model size grow with #GPUs.
Evaluated on 8 AWS EC2 p3.16xlarge nodes with 8 16GB V100s each (64 GPUs in total).

Ablation Study



Combining inter- and intra-operator parallelism scales to more devices.



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)